

(b) (i) Complete the Karnaugh map (K-map) for the Boolean expression:

$$\bar{A}.B.C + \bar{A}.B.\bar{C} + A.\bar{B}.C + A.B.\bar{C}$$

		BC			
A	0	00	01	11	10
	1				

[2]

(ii) Draw loop(s) around appropriate group(s) in the K-map to produce an optimal sum-of-products.

[2]

(iii) Write the Boolean expression from your answer to part **b(ii)** as a simplified sum-of-products. Do **not** carry out any further simplification.

..... [2]

7 (a) Identify **one** Artificial Intelligence (AI) algorithm to find the shortest distance between two points on a graph.

..... [1]

(b) Describe Deep Learning.

..... [5]

